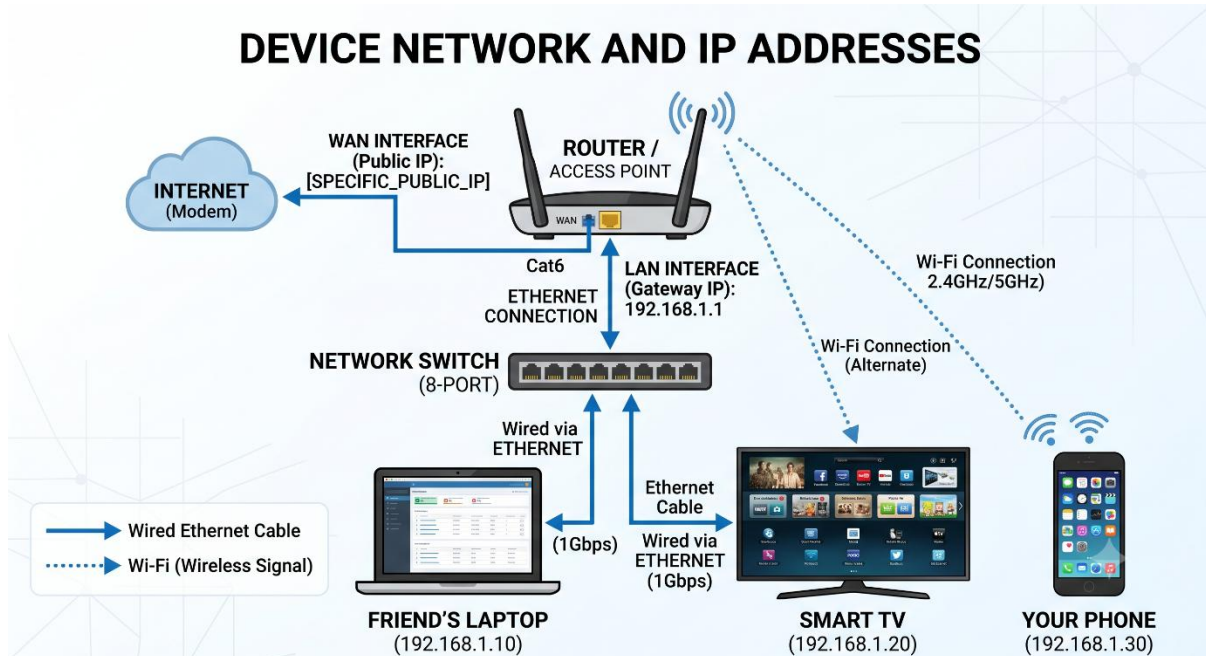


IP routing and routing protocols

Task (1):



Task (2):

Manual IP routing simulation: laptop → friend's computer (different subnet)

Setup (made-up addresses):

- Your laptop: 192.168.1.10/24, gateway 192.168.1.1
- Your friend's computer: 192.168.2.20/24, gateway 192.168.2.1
- Router R1 connects subnet 192.168.1.0/24 (interface eth0) and links to Router R2
- Router R2 connects to subnet 192.168.2.0/24 (interface eth1) where your friend's PC sits

Task (3):

Fictional Zomato-Style Delivery Network Routing Table:

Assume a fictional food-delivery company has three city branches. Each branch uses a different private subnet and connects through a central router.

Branch / City	Network Address	Subnet Mask	Next Hop	Interface
Ahmedabad	10.10.1.0	255.255.255.0 (/24)	Directly Connected	GigabitEthernet 0/0
Mumbai	10.10.2.0	255.255.255.0 (/24)	10.0.0.2	GigabitEthernet 0/1
Delhi	10.10.3.0	255.255.255.0 (/24)	10.0.0.3	GigabitEthernet 0/2

Examples:

Central Router

/ | \

/ | \

Ahm. Mumbai Delhi

10.10.1.0 10.10.2.0 10.10.3.0

/24 /24 /24

Task (4):

Static Routing vs Dynamic Routing

Feature	Static Routing	Dynamic Routing (RIP/OSPF)
Pros	1. Simple for small networks – easy to configure a few routes. 2. Low overhead – does not exchange routing information continuously.	1. Automatically updates routes when the network changes. 2. Scales well for large networks with many branches/servers. 3. Can find alternative paths when a link fails.
Cons	1. Manual configuration is required for every route. 2. Does not automatically adapt to failed links.	1. More complex to configure and troubleshoot. 2. Uses CPU, memory, and bandwidth for routing updates/calculations.
Real-world example	A small restaurant might use a static route between its POS system and local server because the network rarely changes.	A food-delivery service such as Zomato could use OSPF in a large data-center/branch network so traffic can automatically use another route if a network link fails.

Task (5):

Sample OSPF Configuration Between Two Routers

Below is a Cisco IOS example. Assume Router 1 and Router 2 are connected through **10.0.0.0/30**.

Router 1 Configuration:

```
Router1(config)# interface GigabitEthernet0/0
```

```
Router1(config-if)# ip address 10.0.0.1 255.255.255.252
```

```
Router1(config-if)# no shutdown
```

```
Router1(config)# router ospf 1
```

```
Router1(config-router)# router-id 1.1.1.1
```

```
Router1(config-router)# network 10.0.0.0 0.0.0.3 area 0
```

Router 2 Configuration:

```
Router2(config)# interface GigabitEthernet0/0
```

```
Router2(config-if)# ip address 10.0.0.2 255.255.255.252
```

```
Router2(config-if)# no shutdown
```

```
Router2(config)# router ospf 1
```

```
Router2(config-router)# router-id 2.2.2.2
```

```
Router2(config-router)# network 10.0.0.0 0.0.0.3 area 0
```